

Proof of conjecture 00000001012: the subdegrees of $\text{PSL}(n, q)$ on $\text{PG}(n - 1, q)$ are 1 and $(q^n - q)/(q - 1)$

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1 The conjecture and its reading

Statement (English, verbatim). Definition: The permutation representation of minimal degree of classical groups (acting on the points of projective space). Conjecture: The subdegrees of $\text{PSL}(n, q)$ acting on $\text{PG}(n - 1, q)$ are of the explicit $(q^n - q)/(q - 1)$ type (a subdegree formula). (The Chinese text states the same.)

Reading. The subdegrees of a transitive permutation group are the lengths of the orbits of a point stabilizer. Every transitive action has the trivial subdegree 1 (the orbit $\{p\}$ of the fixed point p itself), so “the subdegrees are of the $(q^n - q)/(q - 1)$ type” is read as: the subdegrees are 1 and $(q^n - q)/(q - 1)$, i.e. the only nontrivial subdegree is $(q^n - q)/(q - 1)$. We prove the complete statement: for every point p the stabilizer has *exactly two* orbits, $\{p\}$ and the set of all other points, of sizes 1 and $(q^n - q)/(q - 1)$. This covers both the “full multiset of subdegrees is $[1, (q^n - q)/(q - 1)]$ ” reading and the “nontrivial subdegree equals $(q^n - q)/(q - 1)$ ” reading. The reading in which *every* subdegree, including the trivial one, equals $(q^n - q)/(q - 1)$ is false for every transitive action on more than one point (the orbit $\{p\}$ has size $1 \neq (q^n - q)/(q - 1)$, which is at least 2), and we do not adopt it.

Scope. We take $n \geq 2$ (for $n = 1$, $\text{PG}(0, q)$ is a single point, which we also prove, and there is no nontrivial subdegree) and q the order of an arbitrary finite field F . The Definition line’s phrase “of minimal degree” is not used: the conjecture names the action of $\text{PSL}(n, q)$ on the points of $\text{PG}(n - 1, q)$ explicitly, and that is the action we treat. We do not prove that this action has minimal degree among faithful transitive actions.

2 Definitions (as formalized)

Definition 1. Let F be a finite field with $q = |F|$ elements and $n \geq 0$. $\text{SL}(n, F)$ is the group of $n \times n$ matrices over F of determinant 1 (Mathlib `Matrix.SpecialLinearGroup (Fin n) F`). $\text{PSL}(n, F) = \text{SL}(n, F)/Z(\text{SL}(n, F))$, the quotient by the centre (Mathlib `Matrix.ProjectiveSpecialLinearGroup (Fin n) F`; Lean abbreviation `PSL n F`).

Definition 2. The point set of $\text{PG}(n - 1, F)$ is the set of one-dimensional subspaces of F^n , i.e. nonzero vectors of F^n modulo nonzero scalars (Mathlib `Projectivization F (Fin n -> F)`; Lean abbreviation `PG n F`). $\text{SL}(n, F)$ acts by $g \cdot [v] = [gv]$, and $\text{PSL}(n, F)$ acts through $\text{SL}(n, F)$: Mathlib defines the action of PSL as the factorization of the SL permutation representation through SL/Z ,

and `Matrix.ProjectiveSpecialLinearGroup.smul_proj_mk` states $\bar{g} \cdot x = g \cdot x$ for the image $\bar{g}$ of $g \in \text{SL}(n, F)$ (our `psl_smul_eq`).

Definition 3. For a point p , $\text{Stab}(p) = \{g \in \text{PSL}(n, F) : g \cdot p = p\}$ (Mathlib `MulAction.stabilizer`), acting on the points by restriction. The orbit of x under $\text{Stab}(p)$ is $\text{Stab}(p) \cdot x = \{h \cdot x : h \in \text{Stab}(p)\}$ (Mathlib `MulAction.orbit`); the set of orbits is the quotient `MulAction.orbitRel.Quotient`. The *subdegrees* are the cardinalities `Nat.card` of these orbits.

Conventions. Natural-number arithmetic: $q^n - q$ and $q - 1$ are truncated differences and $/$ is floor division in $\mathbb{N}$; we prove that the division is exact, $(q - 1) \cdot \frac{q^n - q}{q - 1} = q^n - q$, so the formula has its usual meaning.

3 Proof

Lemma 4 (2-transitivity; `psl_two_transitive`). *If $a \neq b$ and $c \neq d$ are points of $\text{PG}(n - 1, F)$, there is $g \in \text{PSL}(n, F)$ with $g \cdot a = c$ and $g \cdot b = d$.*

Proof. Mathlib proves that $\text{SL}(n, F)$ acts 2-transitively on the points (instance `IsMultiplyPretransitive (Matrix.SpecialLinearGroup (Fin n) F) (P F (Fin n) -> F)` 2, derived from `Projectivization.specialLinearGroup_is_two_pretransitive`, file `Mathlib/LinearAlgebra/Projectivization/Action.lean`, by D. Loeffler and A. Chambert-Loir): a linear isomorphism carrying representatives of a, b to representatives of c, d exists because two distinct points have linearly independent representatives, and its determinant is corrected to 1 by a dilatation fixing a and b . By `MulAction.is_two_pretransitive_iff` this gives $g \in \text{SL}(n, F)$ with $g \cdot a = c$, $g \cdot b = d$, and its image $\bar{g} \in \text{PSL}(n, F)$ acts the same way. $\square$

Lemma 5 (`orbit_stabilizer_self`, `orbit_stabilizer_of_ne`). $\text{Stab}(p) \cdot p = \{p\}$, and for $x \neq p$, $\text{Stab}(p) \cdot x = \text{PG}(n - 1, F) \setminus \{p\}$.

Proof. The first is the definition of the stabilizer. For $x \neq p$: if $h \in \text{Stab}(p)$ and $h \cdot x = p = h \cdot p$ then $x = p$ by injectivity of h , so $\text{Stab}(p) \cdot x \subseteq \{p\}^c$. Conversely, for $y \neq p$ apply the previous lemma to $(p, x) \mapsto (p, y)$: the element g found fixes p and sends x to y . $\square$

Lemma 6 (point count; `pow_eq_card_points`, `card_points`, `arith`). *Let N be the number of points. Then $q^n = N(q - 1) + 1$, i.e. $N = (q^n - 1)/(q - 1)$. If $n \geq 2$ then $N \geq 3$, $N - 1 = (q^n - q)/(q - 1)$ and $(q - 1) \cdot \frac{q^n - q}{q - 1} = q^n - q$.*

Proof. The count is Mathlib's `Projectivization.card'` and `Projectivization.card''` (nonzero vectors are partitioned into N classes of size $q - 1$), with $|F^n| = q^n$. Write $q = r + 2$. Since $q^n \geq q^2$, $N(r + 1) + 1 \geq (r + 2)^2$ forces $N \geq 3$. Writing $N = M + 1$, $q^n - q = M(r + 1) + (r + 2) - (r + 2) = M(q - 1)$, so $(q^n - q)/(q - 1) = M = N - 1$, with exact division. $\square$

Theorem 7 (main theorem, `Conjecture1012.subdegrees_PSL`). *Let F be a finite field with q elements, $n \geq 2$, and p any point of $\text{PG}(n - 1, F)$. Then*

1. $\text{PSL}(n, F)$ acts transitively and faithfully on the points of $\text{PG}(n - 1, F)$;
2. the set of orbits of $\text{Stab}(p)$ is exactly $\{\{p\}, \{p\}^c\}$, these two sets are different, and the number of orbits (`Nat.card` of the orbit quotient) is 2;
3. $|\text{Stab}(p) \cdot p| = 1$ and $|\text{Stab}(p) \cdot x| = (q^n - q)/(q - 1)$ for every $x \neq p$;

$$4. (q-1) \cdot \frac{q^n - q}{q-1} = q^n - q \text{ and } \frac{q^n - q}{q-1} = \frac{q^n - 1}{q-1} - 1;$$

5. the set of subdegrees $\{|\text{Stab}(p) \cdot x| : x\}$ equals $\{1, (q^n - q)/(q-1)\}$, and $1 \neq (q^n - q)/(q-1)$.

Proof. Transitivity and faithfulness are found by instance search in Mathlib (Mathlib proves the action primitive, which includes transitivity; `FaithfulSMul` holds since the kernel of the SL action is the centre, `Projectivization.SL_mulAction_ker`). Since $N \geq 3$ there is a point $x_0 \neq p$, so $\{p\} \neq \{p\}^c$. Item 2 follows from the orbit lemma, and the orbit quotient injects into sets of points (`orbitRel.Quotient.orbit_injective`) with image $\{\{p\}, \{p\}^c\}$, so it has 2 elements. Item 3: $|\{p\}| = 1$ and $|\{p\}^c| = N - 1 = (q^n - q)/(q-1)$. Item 4 is the counting lemma. Item 5 follows from items 2 and 3, and $(q^n - q)/(q-1) = N - 1 \geq 2$. $\square$

Remark 8 (`card_points_one`). For $n = 1$ the space $\text{PG}(0, F)$ has exactly one point, so the hypothesis $n \geq 2$ is needed for a nontrivial subdegree to exist.

4 Lean formalization and axiom audit

File `lean/Conjecture1012/Basic.lean` (namespace `Conjecture1012`, only import `Mathlib`). Main theorem:

```
theorem subdegrees_PSL (hn : 2 <= n) (p : PG n F) :
  IsPretransitive (PSL n F) (PG n F) /\ FaithfulSMul (PSL n F) (PG n F) /\
  Set.range (fun x : PG n F => orbit (stabilizer (PSL n F) p) x) =
    {{p}, {p}^c} /\
    ({p} : Set (PG n F)) != {p}^c /\
    Nat.card (orbitRel.Quotient (stabilizer (PSL n F) p) (PG n F)) = 2 /\
    Nat.card (orbit (stabilizer (PSL n F) p) p) = 1 /\
    (forall x : PG n F, x != p -> Nat.card (orbit (stabilizer (PSL n F) p) x) =
      (Fintype.card F ^ n - Fintype.card F) / (Fintype.card F - 1)) /\
    (Fintype.card F - 1) *
      ((Fintype.card F ^ n - Fintype.card F) / (Fintype.card F - 1)) =
      Fintype.card F ^ n - Fintype.card F /\
    (Fintype.card F ^ n - Fintype.card F) / (Fintype.card F - 1) =
      (Fintype.card F ^ n - 1) / (Fintype.card F - 1) - 1 /\
    Set.range (fun x : PG n F => Nat.card (orbit (stabilizer (PSL n F) p) x)) =
      {1, (Fintype.card F ^ n - Fintype.card F) / (Fintype.card F - 1)} /\
    1 != (Fintype.card F ^ n - Fintype.card F) / (Fintype.card F - 1)
```

(ASCII rendering; the source uses the Unicode symbols.) Here F is any type with `[Field F]` `[Fintype F]`, `PSL n F` is `Matrix.ProjectiveSpecialLinearGroup (Fin n) F` and `PG n F` is `Projectivization F (Fin n -> F)`. Supporting theorems: `psl_smul_eq`, `psl_two_transitive`, `orbit_stabilizer_self`, `orbit_stabilizer_of_ne`, `pow_eq_card_points`, `arith`, `card_points`, `card_compl_point`, `subdegree_ge_two`, `card_points_one`.

Axiom audit. `lake build` succeeds, and `#print axioms Conjecture1012.subdegrees_PSL` and `#print axioms Conjecture1012.card_points_one` both report `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

Credit. The 2-transitivity of SL on projective space, the PSL action, its faithfulness and the point count are Mathlib results (`Mathlib/LinearAlgebra/Projectivization/Action.lean` and `Cardinality.lean`).